

MONICA PURUSHOTHAMAN

Front-End Engineer | React | JavaScript | Component Libraries | Responsive & Interactive UI
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PROFESSIONAL SUMMARY

Front-End Engineer with **9+ years** of building and maintaining reusable **React component libraries** and **responsive design-to-code UI** across enterprise and independent products. Proven track record of leading **frontend architecture**. Working command of **accessibility (WCAG)**, **SEO**, page load, rendering, performance optimization, **cross-browser** and multi-device compatibility. Translates designer wireframes and prototypes into stable, scalable UIs backed by unit, integration, and end-to-end tests. Collaborated seamlessly with UX designers, product managers, and backend engineers.

TECHNICAL SKILLS

Frontend: Angular, **React**, **JavaScript** (ES6+), **TypeScript**, **HTML5**, **CSS3**, **SASS**, **JWT**, **RBAC**

Backend & Cloud: Node.js, Express.js, MongoDB, **AWS** (EC2), MySQL 5.7, REST APIs

Libraries & Tools: Cornerstone.js, Hammer.js, Raphael.js, jqPlot, Datatables, Chart.js

DevOps & Testing: Jenkins, SonarQube, **GitHub**, SVN, Selenium, TestNG, **Vercel**, **Render**

Methodologies: MVC Architecture, Agile/Scrum, **Reusable Component Libraries**, **Design-to-Code** Implementation, Cross-browser Compatibility, **Responsive Design**, **SEO**

AI-Assisted Development: Agentic coding tools (**Claude Code**, **GitHub Copilot**) and enthusiastic AI learner.

WORK EXPERIENCE

Front-End Engineer

Independent Developer

Newbury Park, California

2021 – Present

- Continued professional development through 6+ years of self-directed learning and hands-on software development. Built and deployed production full-stack applications applying modern **React hooks**, **component architecture**, RESTful API integration, and **responsive CSS3 layouts** to stay current with industry standards.

Live apps deployed:

- Expense Tracker** — Designed and built a scroll-driven marketing site backed by a fully working expense-tracking app (**Next.js**, **Tailwind**, **GSAP ScrollTrigger**), with pinned storytelling, live data entry, and category-based spend breakdowns. Architected a **reusable component library** on a token-based design system from interaction design to production - <https://expense-tracker-monica.vercel.app/>
- MERN Full-Stack Job Portal** (React, Node.js, Express, MongoDB) – react-job-portal-delta.vercel.app
- Partner Ingestion Console** - that simulates ingesting media partner XML feeds, validates them against a real schema, and tracks onboarding progress (**React**, **Node.js**, **SQL**) – partner-console-app.vercel.app

Senior Frontend Developer

Bank of America Continuum India Pvt Ltd

Chennai, India

2019 – 2020

Quartz UI – Desktop to Web Migration

- Spearheaded the migration of Quartz UI from a Windows Desktop application to a modern Web UI, **improving average page load speed by 38%**.
- Reduced operational costs by 26% by redesigning data exchange and information distribution systems.
- Built a reusable **React** and **JavaScript** component **library**, **cutting project workload by 35%** across multiple teams.
- Facilitated quick access to the records bureau by streamlining navigation and data retrieval flows.

Senior Frontend Developer

Samsung R&D Institute

Bangalore, India

2016 – 2018

HME – IOTA-ADNEX (Health & Medical AI)

- Designed and developed a clinical UI console enabling physicians to assess malignancy probability of ovarian tumors from DICOM scan inputs.
- Improved data visualization and clinical decision-making by 43%** by building complex bar and radar charts to illustrate benign vs. malignant tumor probability.
- Architected the full frontend** using MVC model in **React**, subsequently re-implemented the app in Angular for dual-platform delivery.
- Mentored junior developers**, resulting in a **37% reduction in bug** reports during Angular implementation.

- Received **SPOT AWARD (2018)** for efficient delivery of the IOTA-ADNEX application.

Senior Frontend Developer

Samsung R&D Institute

Bangalore, India

2016 – 2017

HME – Remote Rendering (Fetal Imaging)

- Designed and implemented Frontend architecture to display interactive 2D and 3D fetal images with annotation and image rendering capabilities using the Cornerstone.js library.
- Boosted user satisfaction scores by 38%** through an enhanced interactive experience with customizable presets via **HTML5 Canvas**.
- Modularized the application for full **cross-browser compatibility** on mobile devices using Hammer.js.
- Optimized MySQL database queries, **improving data retrieval time by 34%**.

Frontend Developer

Samsung R&D Institute

Bangalore, India

2015 – 2016

CT Service – Config & Diagnostics UI

- Built the Frontend UI for Config, Diagnostics, Calibration, and Reports modules for CT machine management.
- Developed a **custom JavaScript pagination library** that **reduced medical repository load time by 43%** and increased user engagement by 26%.
- Created a **custom graph library** for visualizing diagnostics and calibration parameters; used Cornerstone.js for ROI annotation on DICOM images.
- Received **EMPLOYEE OF THE MONTH and SPOT AWARD (2016)** for **outstanding delivery**.

Frontend Developer

Samsung R&D Institute

Bangalore, India

2014 – 2015

Techwin – Security Camera Platform

- Developed a Web 'Open App' to streamline firmware installation and uninstallation on security camera systems, **boosting operational efficiency by 65%**.
- Built comprehensive health metrics for uploaded apps, giving users real-time insights that led to a **36% reduction in bugs**.
- Developed a Motion Detection app supporting Play, Delete, Start, and Stop video actions.

Software Programmer

Samsung R&D Institute

Bangalore, India

2013 – 2014

vCDN – Virtual Content Delivery Network

- Built a virtual CDN integrating vNodes and client nodes on Amazon EC2.
- Implemented a centralized controller that **increased network performance by 43%** and enabled robust scalability.

Software Programmer

Cognizant Technology Solutions

Chennai, India

2011 – 2012

Sears E-Commerce – Automation Testing

- Developed a **regression automation tool** using Selenium, TestNG, Jenkins, Java, HTML5, JavaScript, and jQuery for Sears, Kmart, and My Gofer websites.

Software Programmer

Cognizant Technology Solutions

Chennai, India

2010 – 2011

Walmart – Cart & Checkout Optimization

- Contributed to migration from Struts to Spring Web Flow framework; handled maintenance and bug fixing of checkout modules using Java, **HTML5, JavaScript, CSS**, and jQuery.

EDUCATION

SASTRA Deemed University

M.S. in Data Science – In Progress, **Expected Dec 2026**

Online

2025 – 2026

PSG College of Technology

B.E. in Computer Science – GPA: 8.6/10.0

Coimbatore, India

2006 – 2010

WORK AUTHORIZATION

Authorized to work in the United States of America – H4 EAD (Employment Authorization Document).

Visa sponsorship is not required.